

COMPREHENSIVE CANCER CENTER

Division of Medical Oncology – Genitourinary Tumors

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Date: July 25, 1996

To: Dr. Margaret Kellerman, Primary Care Physician
Internal Medicine Associates, 522 Commonwealth Avenue, Boston, MA 02215

From: Dr. Jonathan R. Whitmore, MD, FACP
Medical Oncology – Genitourinary Malignancies

Re: **HICKLE, Brady998**

DOB: November 27, 1914 (Age 81)

MRN: 19141127-GU-0982

Dear Dr. Kellerman,

Thank you for your continued co-management of our mutual patient, Mr. Brady Hickle. I am writing to outline the medical oncology treatment plan we have devised for his recently diagnosed metastatic castration-sensitive prostate adenocarcinoma. Given his complex medical history—including recent cardiac arrest, stroke with residual monoparesis, metabolic syndrome, diabetes, and chronic kidney disease—we have carefully considered both the oncologic imperative and cardiac risk mitigation in formulating this regimen.

Indication and Disease Summary

Mr. Hickle was initially found to have localized prostate carcinoma in situ approximately 18 months ago, managed conservatively given his age and comorbidities. Unfortunately, serial PSA monitoring revealed a rapid rise from 4.2 ng/mL in January 1996 to 38.6 ng/mL in June 1996. Subsequent staging workup, including bone scintigraphy and CT imaging of the abdomen and pelvis completed on July 18, 1996, demonstrated multiple sclerotic osseous lesions in the lumbar spine, bilateral iliac bones, and proximal femora, consistent with diffuse skeletal metastases. Pelvic imaging also noted extracapsular extension and bilateral pelvic lymphadenopathy.

Diagnosis: Metastatic castration-sensitive prostate adenocarcinoma, Gleason score 4+4=8 (high-grade), de novo stage IV, with extensive skeletal involvement.

Treatment Plan

After extensive discussion with Mr. Hickle and his family regarding prognosis, treatment goals, and potential toxicities, we have elected to proceed with combination androgen deprivation therapy (ADT) plus systemic chemotherapy, an approach supported by the CHAARTED and STAMPEDE trial data showing improved overall survival in metastatic hormone-sensitive disease.

Regimen Details:

- **Leuprolide Acetate 22.5 mg intramuscular injection**, administered every 3 months (long-acting depot formulation). First dose administered July 22, 1996.
- **Docetaxel 75 mg/m² intravenous infusion**, given every 3 weeks for a total of 6 cycles (18-week course). BSA calculated at 1.91 m², yielding an absolute dose of 143 mg per cycle. Cycle 1 Day 1 scheduled for July 29, 1996.
- **Dexamethasone 8 mg oral premedication**, administered 12 hours prior, immediately prior, and 12 hours post-docetaxel infusion to mitigate hypersensitivity and fluid retention.
- **Standard antiemetic prophylaxis**: ondansetron 16 mg IV prior to chemotherapy.

Pre-Treatment Laboratory Requirements:

Prior to each cycle of docetaxel, the following laboratory parameters must be met:

Parameter	Required Threshold
Absolute Neutrophil Count (ANC)	$\geq 1,500$ cells/ μ L
Platelet Count	$\geq 100,000$ cells/ μ L
Hemoglobin	≥ 9.0 g/dL
Serum Creatinine	≤ 1.8 mg/dL or CrCl ≥ 40 mL/min
AST / ALT	$\leq 2.5 \times$ ULN
Total Bilirubin	$\leq 1.5 \times$ ULN

Our infusion center will coordinate pre-cycle labs approximately 48–72 hours prior to each scheduled treatment.

Expected Toxicities and Management

Patients receiving docetaxel commonly experience the following adverse effects, and we have instituted appropriate prophylactic and monitoring measures:

- **Myelosuppression**: Neutropenia (nadir typically day 7–10) with associated infection risk. We will monitor CBC with differential prior to each cycle. G-CSF (filgrastim) support may be instituted if febrile neutropenia occurs or ANC recovery is delayed.
- **Fluid retention**: Peripheral edema, pleural effusions. Dexamethasone premedication significantly reduces this risk. Daily weights and clinical assessment at each visit.
- **Fatigue**: Cumulative and often dose-limiting. Counsel patient on energy conservation and consider Epoetin alfa continuation (currently receiving for anemia secondary to chronic kidney disease and myelodysplasia).
- **Neuropathy**: Peripheral sensory neuropathy may develop, typically mild with docetaxel. Will assess at each visit using clinical grading.
- **Alopecia**: Expected and reversible.
- **Gastrointestinal**: Nausea, diarrhea, mucositis—managed with supportive care and dose modification if grade ≥ 3 .
- **Dermatologic**: Nail changes, rash—generally mild.

Mr. Hickle and his family have been counseled on the signs and symptoms of neutropenic fever (temperature $\geq 100.4^{\circ}\text{F}$ with ANC < 500), and instructed to contact our triage service immediately or proceed to the emergency department for urgent evaluation and empiric broad-spectrum antibiotics.

Cardiac Considerations and Risk Mitigation

As you are aware, Mr. Hickle suffered a cardiac arrest on August 9, 1996, following several weeks of declining functional status in the setting of advanced coronary artery disease, prior stroke, and metabolic derangements. He underwent successful resuscitation and has since had a biventricular implantable cardioverter-defibrillator (ICD) placed. Post-arrest echocardiography revealed an ejection fraction of 32%, consistent with ischemic cardiomyopathy.

Anthracycline Contraindication: Given his severely reduced left ventricular function, anthracycline-based chemotherapy (e.g., doxorubicin) is absolutely contraindicated due to unacceptable cardiotoxicity risk.

Docetaxel Cardiac Safety Profile: In contrast to anthracyclines, docetaxel does not exhibit direct myocardial toxicity and is considered safe in patients with pre-existing cardiomyopathy. Fluid retention remains the primary cardiovascular concern, mitigated effectively with corticosteroid prophylaxis. We will coordinate closely with his cardiologist, Dr. Anwar Rashid, and monitor for signs of volume overload, adjusting diuretic therapy as needed.

Baseline ECG and repeat echocardiography are not routinely required for docetaxel, but clinical vigilance for decompensated heart failure will be maintained throughout treatment.

Monitoring and Follow-Up Plan

Mr. Hickle will be seen in our Medical Oncology clinic prior to each chemotherapy cycle (every 3 weeks) for clinical assessment, toxicity evaluation, and laboratory review. Key monitoring elements include:

- **PSA:** Measured every 3 months. Anticipated decline to < 0.2 ng/mL with effective ADT and chemotherapy. Biochemical response is an important prognostic indicator.
- **Testosterone:** Confirm castrate levels (< 50 ng/dL) at 3 months post-initiation of leuprolide.
- **Complete Blood Count:** Prior to each docetaxel cycle and as clinically indicated for infection or bleeding symptoms.
- **Comprehensive Metabolic Panel:** Every 3 weeks (pre-chemotherapy) to monitor renal and hepatic function.
- **Bone health:** Baseline DEXA scan and consideration of bisphosphonate or denosumab therapy given skeletal metastases and ADT-associated bone loss. Zoledronic acid 4 mg IV every 3 months is planned, contingent on renal function.
- **Restaging imaging:** CT chest/abdomen/pelvis and bone scan after completion of 6 cycles of docetaxel (approximately 18 weeks), or sooner if clinical progression is suspected.

Coordination of Care

Given the complexity of Mr. Hickle's medical condition, multidisciplinary coordination is essential. I have communicated this plan with cardiology (Dr. Rashid), nephrology (Dr. Paulson), and our palliative care team. Please do not hesitate to contact me directly should you have questions, concerns, or observe clinical changes requiring urgent reassessment. Our triage nurse line is available 24/7 at (617) 555-2405.

I will send you interim updates after cycle 3 and upon completion of chemotherapy, and I remain available for ongoing co-management discussions as his condition evolves.

Thank you again for entrusting Mr. Hickle's oncologic care to our team. Please feel free to reach out with any questions or concerns.

Sincerely,

Jonathan R. Whitmore, MD, FACP

Medical Oncology – Genitourinary Malignancies

Comprehensive Cancer Center

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CC: Dr. Anwar Rashid (Cardiology), Dr. Leonard Paulson (Nephrology), Oncology Infusion Center (Nursing), Patient Chart